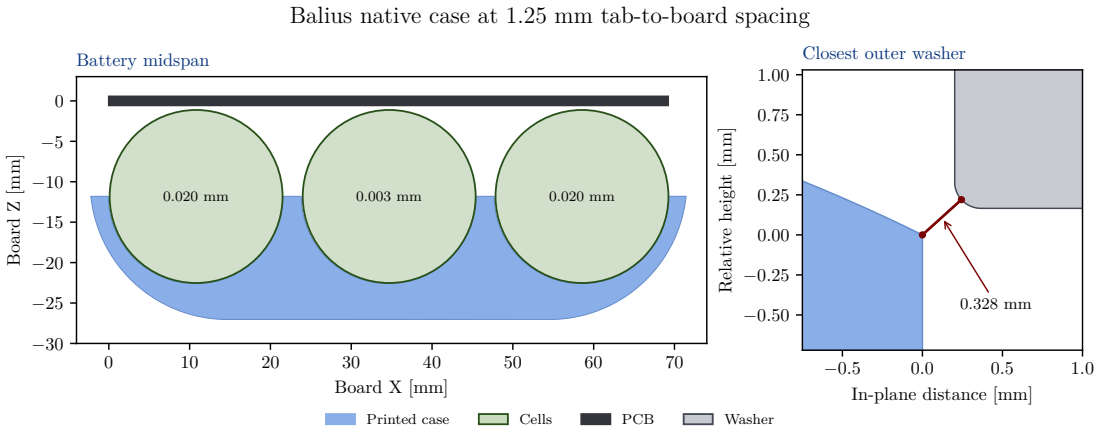
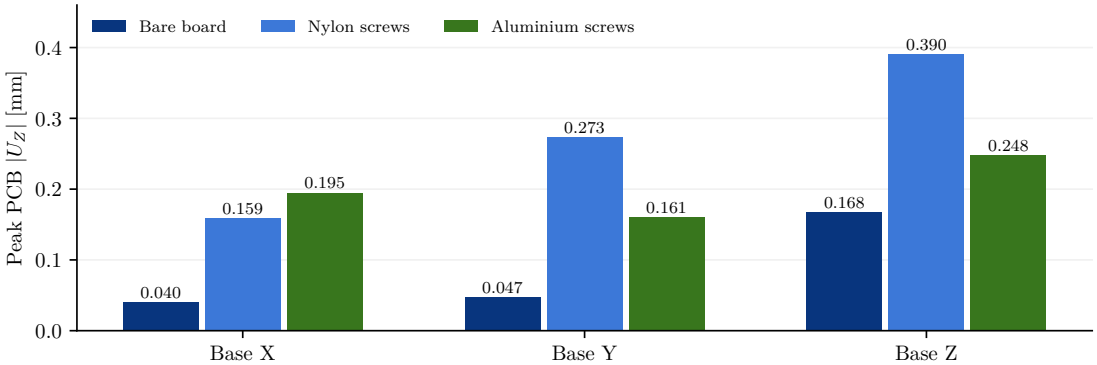
The three-pocket Balius case fits at 1.25 mm stand-off



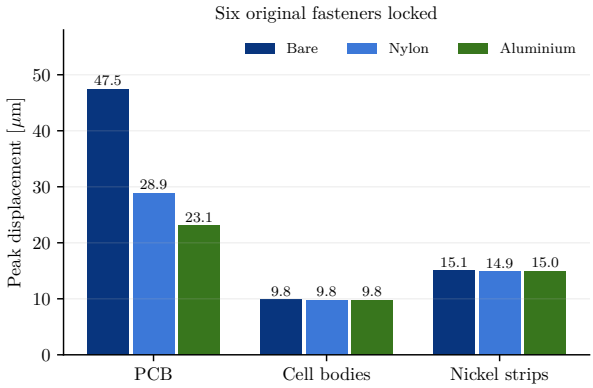
The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. The rigid left cell reaches its pocket after **0.020 mm toward the middle cell** or **0.165 mm away from it**.

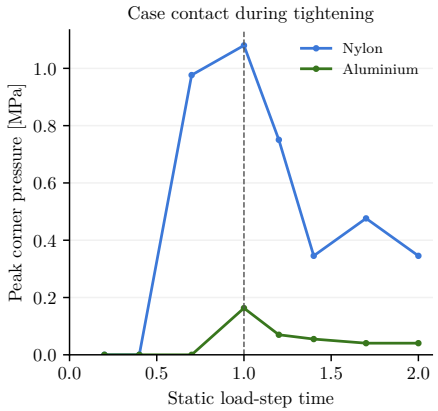
The snug case increases board travel in the seeded clips



During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

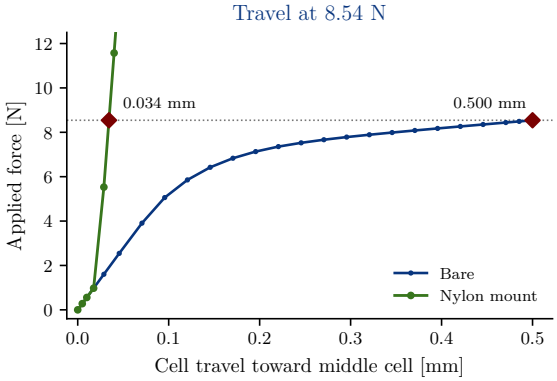
Both snug mounts reduce PCB movement during tightening

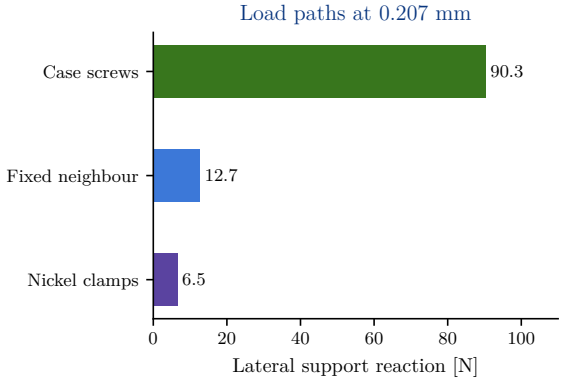




At bolt lock, peak PCB displacement fell from **47.46 μm bare** to **28.91 with nylon screws** and **23.09 with aluminium screws**. All three nylon meshes put the peak at the same corner, with a spread of 0.055 μm. Only the middle cell touched the case.

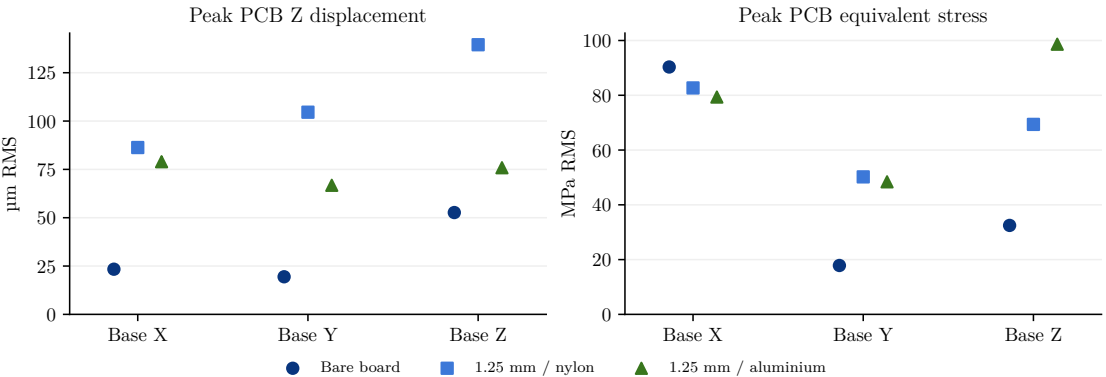
The snug case reduces sideways cell travel at the same load





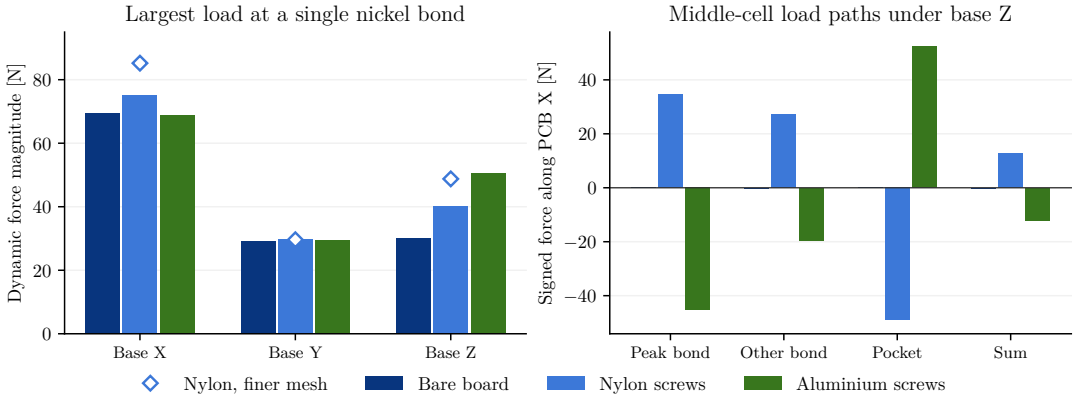
At **8.54 N** toward the middle cell, travel fell from **0.500 mm bare** to **about 0.034 mm with nylon screws**. At 0.265 mm the reaction rises to **1.65 kN**, of which **90% goes into the fixed middle cell**. That restraint makes the high load specific to this fixture.

Aluminium cuts board travel relative to nylon but raises base-Z stress



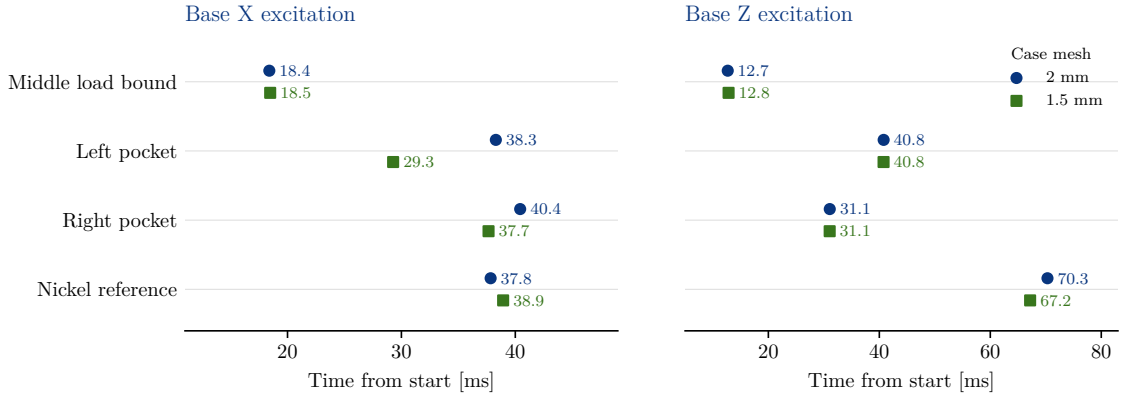
Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

Board-normal vibration loads the middle cell's nickel bond



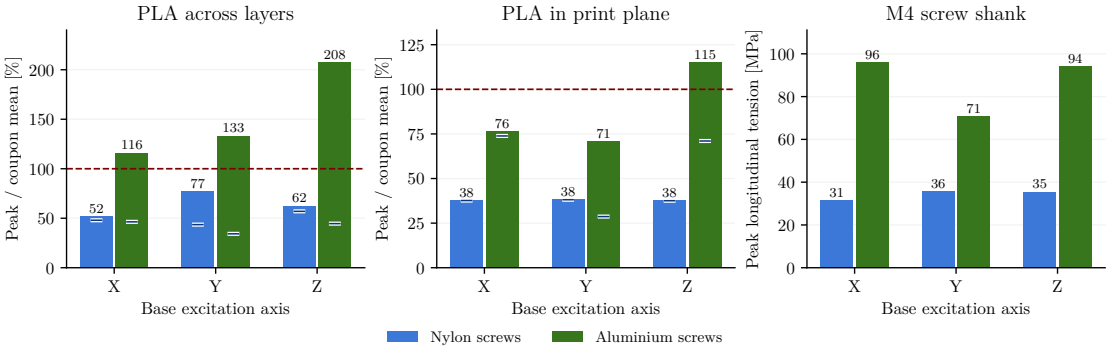
The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The finer nylon mesh gave **48.8 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories at the installed contact state. The diamond uses independent integration on the finer native modes.

The case mesh changes which event comes first under base X



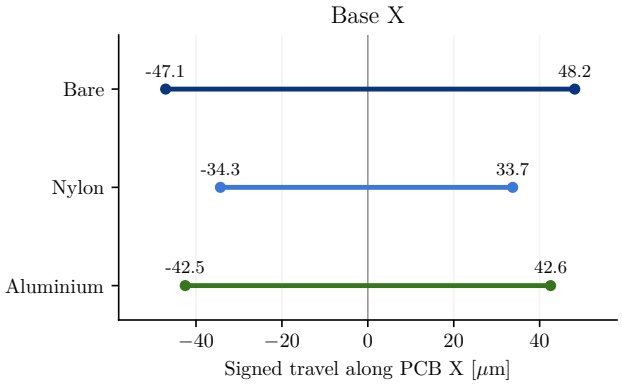
With nylon screws, refining the case moves left-pocket contact from **38.3 to 29.3 ms**. It crosses before the nickel reaches 185 MPa on the fine mesh. The nominal mesh gives the opposite order. Under base Z the right cell comes first on both nylon meshes. The nominal aluminium assembly reaches the left pocket first.

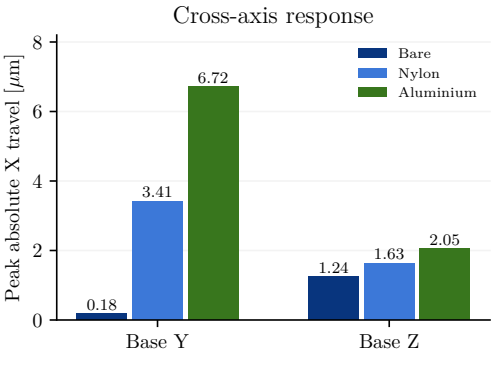
The printed tab exceeds its coupon reference with aluminium screws



On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.

The snug case reduces sideways cell travel under base-X vibration





Relative to its two clamps, the left cell's peak lateral travel fell by **29% with nylon screws** and **12% with aluminium**. Both mounts increased the smaller lateral response under base Y and Z. The comparison uses the nine-second native histories with contact fixed at bolt lock.